

COMPUTER SCIENCE TRIPOS Part IB – 2024 – Paper 7

4 Economics, Law and Ethics (ah793)

market failure

- (a) Explain information asymmetry, and how it can lead to a lemon market.

[6 marks]

Answer: Information asymmetry refers to a situation in which one party involved in a transaction or interaction has more or better information than the other party. In other words, there is an imbalance in the knowledge, data, or information possessed by the participants, leading to a potential disadvantage for the less-informed party (2 marks). Information asymmetry can lead to buyers and sellers having different information about the quality of the products. This can result in a lemon market, in which the buyers do not have knowledge about product quality, and therefore those selling quality products are unable to differentiate from sellers with poor quality products (2 marks). As those selling quality products cannot compete with the low prices, they subsequently leave the market (2 marks).

market failure

- (b) What solutions can help prevent a lemon market?

[6 marks]

Answer: The solution is to signal hidden information (2 marks). Sellers may offer signals about the trustworthiness of their products, such as by offering a warranty or offering to refund if the product is of poor quality (2 marks). Platforms and marketplaces may provide signals through reputation and recommender systems, whereby customers can provide reviews and ratings relating to products/sellers (2 marks).

contemporary
ethical issues

- (c) How is information asymmetry relevant to privacy?

[8 marks]

Answer: There is considerable information asymmetry when it comes to large tech companies, who hold lots of personal information about users, but very little is known about them (2 marks). Users may not know about how their data is processed and analysed, leading to an information imbalance. Individuals may not know the extent to which their personal information is being used or shared with third parties, such as advertisers or analytics providers, or what is being inferred (2 marks). They may have limited control over their data (2 marks). Other areas relevant to privacy and information asymmetry in relation to computer science include security breaches, as individuals may not know their data has been breached, by who, and government surveillance, where individuals may not be aware of the extent and nature of surveillance activities (2 marks).
